American Computer Science League

2023-2024 ● Contest 1: Short Problems Solutions ● Classroom Division

1. Computer Number Systems

$2324_{10} = 4 * 8^3 + 4 * 8^2 + 2 * 8^1 + 4 * 8^0$
$= 4 * 512 + 4 * 64 + 2 * 8 + 4$
$= 4424_8$

D. 4424

2. Computer Number Systems

$73452_8 - 4261_8 + 752_8$
$= (73452_8 - 4261_8) + 752_8$
$= 67171_8 + 752_8$
$= 70143_8$
$70143_8 = 111\ 000\ 001\ 100\ 011_2$
$= 111\ 0000\ 0110\ 0011_2$
$= 7\ \ 0\ \ 6\ \ 3_{16}$

C. 7063

3. **Recursive Functions**

$f(22) = f(22 - 4) + 1 = f(18) + 1 = 17 + 1 = 18$
$f(18) = f(18 - 4) + 1 = f(14) + 1 = 16 + 1 = 17$
$f(14) = f(14 - 4) + 1 = f(10) + 1 = 15 + 1 = 16$
$f(10) = 2 * f(10 + 1) - 3 = 2 * f(11) - 3 = 2 * 9 - 3 = 15$
$f(11) = 2 * f(11 + 1) - 3 = 2 * f(12) - 3 = 2 * 6 - 3 = 9$
$/f(12) = f(12 - 4) + 1 = f(8) + 1 = 5 + 1 = 6$
$f(8) = 8 - 3 = 5$ Now substitute backwards.

E. 18

4. **Recursive Functions**

$f(24) = f(24/2) + 4 = f(12) + 4 = 15 + 4 = 19$
$f(12) = f(12/2) + 4 = f(6) + 4 = 11 + 4 = 15$
$f(6) = f(6/2) + 4 = f(3) + 4 = 7 + 4 = 11$
$f(3) = f(3 - 1) + 2 = f(2) + 4 = 5 + 2 = 7$
$f(2) = f(2/2) + 4 = f(1) + 4 = 1 + 4 = 5$
$f(1) = 1$
So $f(f(24)) = f(19) = 15$
$f(19) = f(19 - 1) + 2 = f(18) + 2 = 13 + 2 = 15$
$f(18) = f(18/2) + 4 = f(9) + 4 = 9 + 4 = 13$
$f(9) = 9$

C. 15

5. What Does This Program Do? (Branching)

Given the rules to follow, only the amount over the amount specified is given the greatest percent. The other categories are paid 3% of the second $5000 and 1% of the first $5000 which is $50. The results are:

a) the amount over a certain threshold is never used
b) this works correctly though not as efficiently designed
c) this is not as well written, but also works
d) this works correctly by using the opposite order for the if…else if…else

Therefore, the answer is b, c, d.

A. b, c, d

6. Computer Number Systems

$2023_{10} = 3747_8$
3747, 3750, …, 3757, 3760, … 3767, 3771, 3772, 3773 is a palindrome.
$3773_8 = 3 * 8\text{^}3 + 7 * 8\text{^}2 + 7 * 8 + 3 = 3 * 512 + 7 * 64 + 56 + 3 = 2043_{10}$

D. 2043

7. Computer Number Systems

$F43A6C3091_{16} = 1111\ 0100\ 0011\ 1010\ 0110\ 1100\ 0011\ 0000\ 1001\ 0001_2$
$= 1\ 111\ 010\ 000\ 111\ 010\ 011\ 011\ 000\ 011\ 000\ 010\ 010\ 001_2$
$= 1\ \ 7\ \ 2\ \ 0\ \ 7\ \ 2\ \ 3\ \ 3\ \ 0\ \ 3\ \ 0\ \ 2\ \ 2\ \ 1_8$

There are 3 0's, 2 1's, 4 2's, 3 3's, 0 4's, 0 5's, 0 6's, and 2 7's. A 2 occurs most.

C. 2

8. Recursive Functions

$f(17, 11) = f(11 - 1, 17 - 2) + 1 = f(10, 15) + 1 = 15 + 1 = 16$
$f(10, 15) = f(10 - 1, 15 - 2) + 3 = f(9, 13) + 3 = 12 + 3 = 15$
$f(9, 13) = f(9 - 1, 13 - 2) + 3 = f(8, 11) + 3 = 9 + 3 = 12$
$f(8, 11) = f(8 - 1, 11 - 2) + 3 = f(7, 9) + 3 = 6 + 3 = 9$
$f(7, 9) = f(7 - 1, 9 - 2) + 3 = f(6, 7) + 3 = 3 + 3 = 6$
$f(6, 7) = f(6 - 1, 7 - 2) + 3 = f(5, 5) + 3 = 0 + 3 = 3$
$f(5, 5) = 5 - 5 = 0$

E. 16

9. Recursive Functions $f(24) = 2 * f(24 - 4) = 2 * f(20) = 2 * 60 = 120$ $f(20) = 2 * f(20 - 4) = 2 * f(16) = 2 * 30 = 60$ $f(16) = f(16 + 1) - 1 = f(17) - 1 = 31 - 1 = 30$ $f(17) = f(17 + 1) - 1 = f(18) - 1 = 32 - 1 = 31$ $f(18) = 2 * f(18 - 4) = 2 * f(14) = 2 * 16 = 32$ $f(14) = 14 + 2 = 16$	D. 120
10. What Does This Program Do? The year 2023 is the only year other than 2400 that is the product of the sum of its digits and the square of the sum of the squares of its digits. In other words, (2+0+2+3)*(2^2+0^2+2^2+3^2)^2 = 7*17^2 = 7*289 = 2023. Its prime factors also are these two sums since 7*17*17 is its prime factorization.	C. 2023